JOSEPH ALAN B. VERGARA

Embedded Systems | Firmware | Edge AI | Sensor Telemetry
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Portfolio • LinkedIn • GitHub

Profile

Fourth-year BS Computer Applications student specializing in embedded systems at MSU-IIT. Develops C/C++ firmware, real-time systems, on-device computer vision, and telemetry web applications for academic research and sensor-based monitoring.

Education

- **BS Computer Applications, Major in Embedded Systems (CGPA: 1.96) —** Mindanao State University – Iligan Institute of Technology | August 2023 – Present; fourth year.
- **Senior High School, Accountancy, Business & Management (Academic Honors) —** MSU-IIT Integrated Developmental School | 2021 – 2023.

Research & Development Experience

- **MIOW Learning Platform — Co-Author & Full-Stack / Embedded Developer.** PhD dissertation research collaboration | 2024 – Present. Built a React, Node.js, and PostgreSQL platform for microcontroller telemetry and automated laboratory scoring; benchmarked cloud and ESP32-P4 edge hosting. Research in progress. GitHub · Live demo
- **Microplastic Detection — PWA & Mobile Developer.** Master's and undergraduate research collaboration | 2025 – Present. Built a cross-platform PWA integrating quantized YOLOv8n inference, particle overlays, camera lifecycle management, and Android APK release toolchains. Research in progress. GitHub · Documentation

DepEd science investigatory research: developed web applications and integrated hardware and software for the following competition projects.

- **O.I.N.K. — Firmware & Web App Developer.** Integrated ESP32 firmware and an MLX90640 thermal sensor with OpenCV processing and a Flutter web app for live thermal telemetry, heatmaps, and fever-threshold alerts. Live demo
- **S.L.I.D.E. — Telemetry Web App Developer.** Developed a monitoring PWA integrating ground-tilt, soil-moisture, and rainfall telemetry from field hardware, with offline caching, sensor trend charts, and threshold alerts. GitHub · Live demo
- **ARUGA — Computer Vision & Web App Developer.** Integrated camera hardware with Python, OpenCV, and MediaPipe Pose for fall detection and inactivity monitoring; developed a web app for live visual telemetry and emergency alerts. GitHub · Live demo

Embedded Systems & Edge AI Projects

- **IoT Human Activity Recognition & Edge Telemetry (BCA182 Lab 3) —** Built an STM32/RT-Spark activity-recognition pipeline with IMU feature extraction, MQTT ingestion, and a telemetry dashboard. GitHub
- **Personal MP3 Player & Tone Sequencer —** Implemented a FreeRTOS audio sequencer with LCD rendering, button decoding, ADC volume control, and 18 unit tests. GitHub
- **Multisensor Room Monitor —** Implemented five FreeRTOS tasks for sensor acquisition, OLED output, encoder navigation, logging, and power management. GitHub · Write-up
- **Snake OS: Embedded Interactive HMI Game —** Developed a Renesas RA6M3 arcade game using RT-Thread and LVGL, with AI navigation and persistent high scores. GitHub · Write-up
- **Edge AI Glasses Object Detection —** Trained and quantized a wearable object-detection model using Roboflow datasets and Edge Impulse. Project
- **Industrial Oven Simulation & Control —** Engineered CODESYS ladder logic and an HMI with heating control, safety interlocks, and threshold alarms. GitHub · Documentation
- **IoT Vermicomposting Monitor —** Implemented ESP32 soil-moisture monitoring, closed-loop irrigation, and environmental telemetry over MQTT. Documentation
- **Line Following Robot —** Programmed AVR Assembly firmware and dual PWM timers for autonomous line tracking using a five-channel IR array. Documentation

Additional Engineering & Software Projects

- **Agap AI Emergency Triage Platform —** Developed an emergency-response platform that converts voice reports into structured, triage-ready incident data. Live demo
- **K-Bin: Automated Smart Waste Sorting —** Built a Qt C++ interface and sensor-fusion system for three waste streams, with SQLite maintenance logging. Documentation
- **Adjustable DC Power Supply —** Designed a regulated variable-output supply up to 12 V with short-circuit protection for embedded prototyping. Portfolio
- **UNICHAT — Anonymous Campus 1-on-1 Chat —** Built an installable campus chat PWA gated by verified institutional emails (@g.msuiit.edu.ph), featuring PostgreSQL transactional matchmaking, real-time messaging, and MFA moderation. GitHub · Live demo
- **Pulse — Unified Observability & Telemetry Radar —** Built a Next.js and Neon PostgreSQL analytics platform with client telemetry beacons, live session dashboards, and automated webhook alerts. GitHub · Live demo
- **Vault — Personal Expense & Family Sanctuary PWA —** Developing an offline-first expense tracker and family planner with SMS transaction parsing, budget analytics, PIN protection, and local encryption. In progress. GitHub · Live demo
- **oddJobs — Hyperlocal Student Workforce Marketplace —** Built a Next.js campus marketplace for peer errands and micro-tasks, with task posting, bidding, status tracking, and authenticated profiles. GitHub · Live demo
- **Barangay Connect Web Application —** Built a React and Supabase community portal for incident reports, document requests, polls, and notifications. GitHub · Live demo
- **Sustainability in the Cup Web —** Built a sustainability research site with green-claim evaluation, survey instruments, and data visualization. Live demo

Technical Skills

- **Programming —** C, C++, Python, JavaScript, TypeScript, Bash, AVR Assembly, and PLC ladder logic.
- **Embedded & RTOS —** STM32, ESP32/ESP32-P4, AVR, Renesas RA6M3, FreeRTOS, RT-Thread, PCB layout, soldering, and hardware debugging.
- **Edge AI & Computer Vision —** YOLOv8n, OpenCV, MediaPipe, TensorFlow Lite, Edge Impulse, and model quantization.
- **Web & Development Tools —** React, Next.js, Node.js, Flutter, PostgreSQL/Supabase, Git, Linux, and Android SDK.
- **Interfaces & Protocols —** I2C, SPI, UART, CAN, MQTT, and HTTP/REST.

Languages: English, Filipino, and Cebuano.

Awards & Recognitions

- **IEEE Sumpai Hackathon 2026 —** 3rd Place and Top 5 Finalist; developed Agap AI | September 14, 2026.
- **TechShowcase 2025 —** 2nd Best Innovation and 3rd Best Pitch; developed Barangay Connect | December 17, 2025.

Leadership

- **Undersecretary, Sports & Development Affairs —** College of Computer Studies Executive Council, MSU-IIT | August 2025 – May 2026.
- **Team Captain, Mobile Legends: Bang Bang —** Titans Esports | 2021 – 2023.

Training, Webinars & Professional Development

- **Startup Basics 101 —** iDEYA: Center of Innovation and Technopreneurship, MSU-IIT | September 24, 2026.
- **AI Career Readiness Training —** MSU-IIT | August 20, 2026.
- **Quantum Circuits as Pictures: An Introduction to ZX Calculus —** Webinar | March 20, 2026.
- **Permaculture: “From Code to Crops” —** Webinar | December 22, 2025.
- **my.ComApps Technology Symposium: RT-Thread Based Technology —** MSU-IIT | November 24–25, 2025.
- **Seize Your Opportunity: Internships and OJTs Workshop —** MSU-IIT | November 24, 2025.
- **Software Freedom Day 2023 —** MSU-IIT | September 16, 2023.